

Qatar-1b

R-2705

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160911 · created 2026-09-01T00:41:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457642.7659 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.053
Transit depth (Rp/Rs)^2	1.33 +/- 1.23 [%]
Orbital Inclination (inc)	86.16 +/- 2.82
Airmass coefficient 1 (a1)	0.779 +/- 0.073
Airmass coefficient 2 (a2)	0.165 +/- 0.045
Scatter in the residuals of the lightcurve fit is	9.05 %
Best Comparison Star	#3 - [557, 163]
Optimal Aperture	2.08
Optimal Annulus	6.78
Transit Duration (day)	0.071 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.115 $\pm$ 0.053 vs 0.1463 (deviation 0.36 σ)
Duration fit vs published	0.071 d vs 0.0692 d (deviation 0.08 σ)
Mid-time O-C	-2.1 min
Depth SNR	1.3
Residual scatter	9.05 %

Light curve

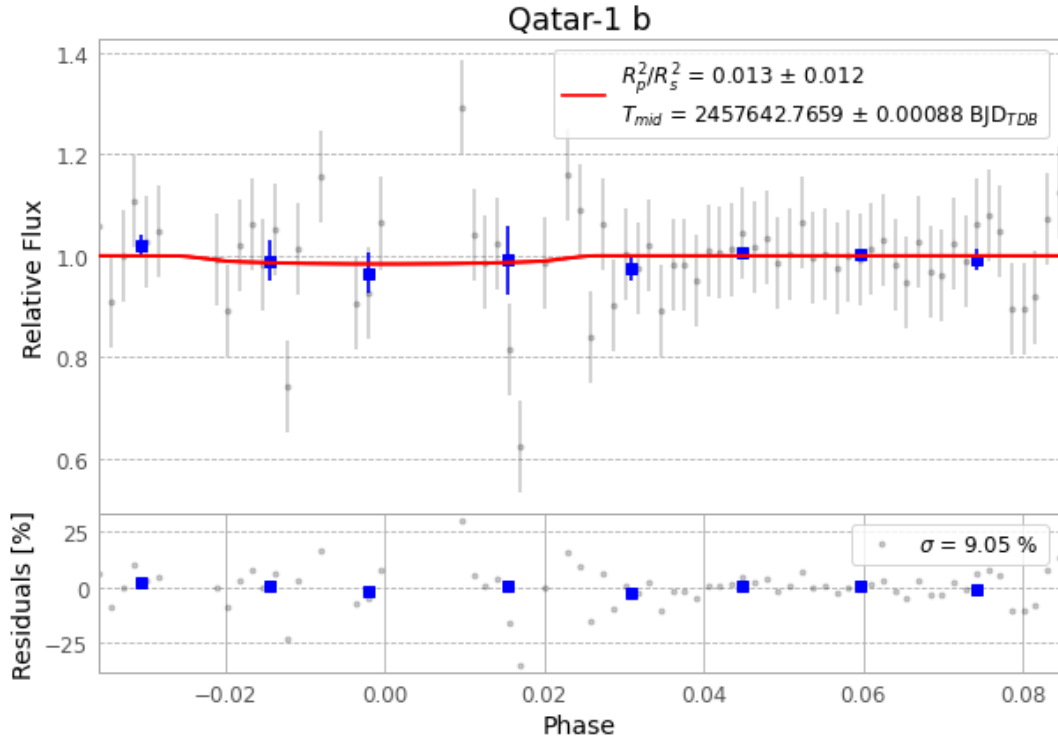


Figure 1 — FinalLightCurve_Qatar-1 b_2016-09-10.png (SHA-256 802a9a5b1533e352...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [200, 292], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 e2004dfaef967f51...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

83 FITS frames (55 MB), Qatar-1160911050612.FITS → Qatar-1160911091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160911.log (319c820e1d94...), FinalLightCurve_Qatar-1 b_2016-09-10.png (802a9a5b1533...), FinalLightCurve_Qatar-1 b_2016-09-10.csv (e76fdbc7fa...)

Compute resources

Wall time 159.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:41Z.